Total Number of Pages: 2			Stream:- B. TECH
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Mid-Term Examination : 2021-22

SUBJECT: MATHEMATICS-II

BRANCH: ALL **Section- ALL**

Answer Question No.1 which is compulsory and any Three from the rest.

The figures in the right hand margin indicate marks.

Q1		Answer the following questions:	(0.5 x 10)
	a)	Give an example of three different matrices such that $AB = AC$ but $B \neq C$.	
	b)	How can you say a system of equation has unique solution, no solution and infinitely many solutions using rank?	
	c)	For what values of t the set of vectors $[2,-4,1],[-2,t,-1],[-1,3,t]$ are linearly dependent and independent?	
	d)	Prove thatany square matrix can be expressed uniquely as the sum of a symmetric and skew-symmetric matrices.	
	e)	Prove that determinant of an orthogonal matrix is ± 1.	
	f)	If λ is an eigen value of a non-singular matrix A, then prove that $\frac{1}{\lambda}$ is an eigen value of A^{-1}.	
	g)	Define algebraic multiplicity and geometric multiplicity.	
	h)	Give an example to show that the eigen values of a symmetric matrix are real.	
	i)	Find the symmetric co-efficient matrix from the quadratic form $(x_1 - x_2)^2 - 4x_3^2$.	
	j)	Find the inverse transformation of $y_1 = 3x_1 + 2x_2$, $y_2 = 4x_1 + x_2$.	
Q2	a)	Find the inverse of the matrix $A = \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$ by using Gauss-Jordan elimination method.	(2.5)
	b)	Find the values of a and b such that the following system of equations has (i) unique solution, (ii) infinitely many solutions and (iii) no solution. $-x+y+2z=2,$ $3x-y+z=6,$ $-x+3y+az=b$	(2.5)

Q3	a)	Evaluate the determinant by reducing to triangular form. $A = \begin{vmatrix} 2 & 0 & -4 & 6 \\ 4 & 5 & 1 & 0 \\ 0 & 2 & 6 & -1 \\ -3 & 8 & 9 & 1 \end{vmatrix}$	(2.5)
	b)	Transform the quadratic form to principal axes form and then find out what conic section it represents. $32x_1^2 - 60x_1x_2 + 7x_2^2 = -52$.	(2.5)
Q4	a)	Define rank of a matrix and find the rank of $\begin{bmatrix} 2 & 0 & 1 & 3 \\ -2 & 4 & 6 & -3 \\ 1 & -4 & 1 & -5 \end{bmatrix}$.	(2.5)
	b)	Diagonalize the matrix $\begin{bmatrix} 1 & 0 & 0 \\ -4 & 2 & 0 \\ 5 & -1 & 3 \end{bmatrix}$.	(2.5)
Q5	a)	Solve the system of equations using Gauss elimination method. $x+2y+3z=0,$ $3x+4y+4z=0,$ $7x+10y+12z=0$	(2.5)
	b)	Find the eigen values and eigen vectors of $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$	(2.5)

Q.No	PART-A (All Questions are compulsory)	
1a	What is an asymptote of a curve? How many asymptotes are there in a parabola?	
1b	State Green's theorem.	
1c	What is the necessary condition for existence of extremum value of a function of two variables f(x, y)? Whether maximum value of every function exists?	
1d	What is the role of an Integrating Factor in the solution of a differential equation? Explain.	
1e	By using convolution find the inverse Laplace transform of $\frac{1}{(s-1)^2}$	
1f	Using first shifting theorem find $L\{e^{-t}\cosh 3t\}$.	
1g	How can you say that two functions are linearly independent? Whether sin x and cosec x are linearly independent?	
1h	Test the convergence of of the series $\sum_{n=2}^{\infty}\frac{1}{\log n}$.	
1i	Find the radius of convergence of the power series $\sum_{m=0}^{\infty}\frac{(-1)^m}{k^m}x^{2m}$.	
1j	Find the directional derivative of the function $f = e^x \cos y$ at the point P(2, π) in the direction of the vector $\vec{b} = i + j$.	
	PART-B (Each Question has internal choices)	
2a	What is an asymptote of a curve? Describe all asymptotes of the curve $y^3 + x^2y + 2xy^2 - y + 1 = 0$.	
2b	Discuss the maximum and minimum values of the function $x^3 + 2xy^2 - 3x^2 - 3y^2 + 4 = 0$.	
	(OR)	
2a	State Taylor's theorem for functions of two variable. Obtain Taylor's formula for the function e^{x+y} at (0, 0) for n=3.	
2b	What is the necessary condition for convergence of a series? Discuss the convergence of the series $\frac{1}{\sqrt{1.2}}+\frac{1}{\sqrt{2.3}}+\frac{1}{\sqrt{3.4}}+\cdots$	
3a	Using the concept of radius of curvature, find the radius of curvature at any point on the curve $x = a(\cos t + t\sin t), y = a(\sin t - t\cos t)$	
3b	Show that the form under the integral sign is exact in space and then integrate $\int_{(0,0,0)}^{(4,1,2)} 3ydx + 3xdy + 2zdz$.	5
	(OR)	
3a	If a wet sheet in a dryer loses its moisture at a rate proportional to its moisture content and it loses half of its moisture during first 10 minutes, when will it be practically dry, say when it will have lost 99% of its moisture?	5
3b	If R=8 ohms, L=2 henrys, C= 0.1 farad, E=160 cos 5t volts, analyse the amount of current flows in the RLC circuit.	5
4a	Find the Fourier series of the function $f(x) = k$ if $-\frac{\pi}{2} < x < \frac{\pi}{2}$ and $f(x) = 0$ if $\frac{\pi}{2} < x < \frac{3\pi}{2}$ where the function is assumed to have period 2π.	
4b	If $P_n(x)$ be the Legendre's polynomial, then how can you prove that $(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x)$	
	(OR)	
4a	Find a power series solution in powers of x of the differential equation $y'' - y = 0$.	
4b	Show that the differential equation $(a^2 - x^2)y'' - 2xy' + 6y = 0$ can be reduced to Legendre equation, then solve it.	
5a	Applying Laplace transform, solve the DE $y'' - 3y' + 2y = 4t - 8,\ y(0) = 2,\ y'(0) = 7$.	
5b	Discuss the value of the integral $\int_0^2\int_{-y}^{y}(x^2 + y^2)dxdy$ by reversing the order.	
	(OR)	
5a	Calculate the amount of work done by the force $\vec{F} = [2z, x, -y]$ by displacing a point along the curve $C: \vec{r} = [\cos t, \sin t, 2t]$ from (1, 0, 0) to (1, 0, 4π).	
5b	Test the exactness of the following differential equation. If exact, solve it.If not use an integrating factor to solve $(3x^2y^4 + 2xy)dx + (2x^3y^3 - x^2)dy = 0$.	
6a	Find $L^{-1}\{\frac{2s+1}{s^2+2s+5}\}$.	
6b	Solve the IVP $x\frac{dy}{dx} + y\log y = xye^x,\ y(1) = 1$.	
	(OR)	
6a	Solve by using method of variation of parameter $y'' + 9y = cosec\ 3x$.	
6b	Solve the integral equation by using Laplace transform $y(t) = \int_0^t y(\tau)\sin 2(t-\tau)d\tau + \sin 2t$.	

.No	PART-A (All Questions are compulsory)	Marks	
1a	Show that the determinant of an orthogonal matrix can take only two values.	2	C
1b	Write down the Simpson's $3/8^{th}$ rule for numerical integration.	2	C
1c	Express the following matrix as the sum of symmetric and skew-symmetric matrices, $\begin{bmatrix} 3 & -4 & -1 \\ 6 & 0 & -1 \\ -3 & 13 & -4 \end{bmatrix}$.	2	C
1d	Find the second order divided difference for the function $f(x)=\frac{1}{x}$.	2	C
1e	How can you say a system of equation has unique solution, no solution and infinitely many solutions using rank?	2	C
1f	Using Euler's method, find $y(0.1)$ if $y'=x^2+y^2$, $y(0)=1$.	2	C
1g	Prove that the set $\{(1,-1,1),(0,1,1),(1,1,1)\}$ is a basis for V_3.	2	C
1h	Find the dimension of the vector space, $V=\{(v_1,v_2,v_3): v_1+v_2-v_3=0\}$.	2	C
1i	Construct forward difference table and then find $\Delta^2 y(10)$ from the following data (10,1.1), (20, 2.0), (30, 4.4), (40, 7.9).	2	C
1j	Evaluate $\int_0^1 (4x-3x^2)\,dx$ by using trapezoidal rule, taking six intervals.	2	C
	PART-B (Each Question has internal choices)		
2a	Define rank of a matrix and find the rank of $\begin{bmatrix} 3 & -1 & 5 \\ 2 & -4 & 6 \\ 10 & 0 & 14 \end{bmatrix}$.	5	
2b	Diagonalize $\begin{bmatrix} -43 & 77 \\ 13 & 93 \end{bmatrix}$ after finding the basis of eigen vectors.	5	
	(OR)		
2c	Design Newton's iteration for cube roots and using this compute the real cube root of 13 correct to four decimal places.	5	
2d	Solve the following system of equations $3x+2y+z=7$, $2x+y+3z=7$, $x+3y+2z=4$ by using Gauss-Seidel iteration method using four significant digits, stating with initial approximation $x=y=z=1$.	5	
3a	Using Lagrange's interpolation formula find the value of $\sqrt{155}$ from the following table of values: x: 150, 152, 154, 156 $y=\sqrt{x}$: 12.247, 12.329, 12.410, 12.490	5	
3b	For what values of λ and μ, the following system of equations posses unique, infinite and no solutions. $2x+3y+5z=9$, $7x+3y-2z=8$, $2x+3y+\lambda z=\mu$.	5	
	(OR)		
3c	Find the inverse of the following matrix by using Gauss-Jordan Elimination method: $\begin{bmatrix} 1 & 8 & -7 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$.	5	
3d	Evaluate $\int_0^{0.4} xe^{-x^2}\,dx$ by using Simpson's $1/3^{rd}$ rule with h = 0.05. Compare your result with the exact value of the integral.	5	
4a	Evaluate the determinant value of following matrix by reducing to its triangular form: $\begin{bmatrix} 2 & 0 & -4 & 6 \\ 4 & 5 & 1 & 0 \\ 0 & 2 & 6 & -1 \\ -3 & 8 & 9 & 1 \end{bmatrix}$.	5	
4b	Using the 4th order Ruge-Kutta method, estimate $y(0.2)$ and $y(0.4)$ for $y'=-0.2xy$, $y(0)=1$, $h=0.2$.	5	
	(OR)		
4c	What type of conic section is represented by the following quadratic form: $-11x_1^2+84x_1x_2+24x_2^2=156$?	5	
4d	Evaluate $f'(1.4)$ by using numerical differentiation with the following data: x: 1.0, 1.2, 1.4, 1.6 $f(x)$: 2.7183, 3.3201, 4.0552, 4.9530	5	
5a	Find a solution of the equation $x^4-x-0.12=0$ by using Regula-falsi method, correct upto three decimal places.	5	
5b	Prove that any orthogonal set of non-zero vectors in an inner product space are linearly independent.	5	

B- decoding

C- opening

D- closing

i) Semantic noise means_____.

A- physical barrier

B- cultural barrier

C- linguistic barrier

D- psychological barrier

j) Efficacy of written communication depends upon_____

A- lengthy presentation

B- complex sentence structure

C- editing after writing

D- brevity in sentences

E- use of strong words

F- effective use of words

Ans.

A- A B and C only

B- B C and D only

C- C D and F only

D- D E and F only

Answer the questions in 50-60 words. Attempt any three.

Q. 2.

What is the importance of the following in Communication?

a) Barriers & Filters 2.5

b) Body Language 2.5

Q.3.

a) Has the development of internet and social media caused a change in the way we communicate (quality, quantity, style etc.)?2.5

b) What is the contribution of non-verbal communication in the process of communication? 2.5

Q.4.

a) Describe the role of channel in the process of Communication. 2.5

b) Explain how silence works as a tool for Communication. 2.5

Q. 5.

a) Which principle of effective communication is used at the beginning of Communication? 2.5

b) What do you understand by a Communication loop? Explain with the help of a diagram. 2.5